

NUTRIAI

AI-Powered Clinical Nutrition for Nigeria

Project Overview

NutriAI is a mobile-first clinical nutrition application that uses artificial intelligence to deliver personalized, culturally-relevant meal planning and dietary guidance for Nigerians. Unlike global nutrition apps that treat Nigerian cuisine as an afterthought, NutriAI is built from the ground up around the foods Nigerians actually eat — egusi, pounded yam, suya, moi moi, jollof rice — and the health conditions they actually face.

The app integrates four intelligence layers:

1. **Nigerian Food Intelligence Engine** — A curated database of 2,000+ local foods with verified macronutrient, micronutrient, and glycemic profiles
 2. **Clinical Nutrition Protocols** — Evidence-based dietary guidance for PCOS, diabetes, hypertension, pregnancy, and weight management
 3. **Cycle-Synced Nutrition** — The first African nutrition app to adapt meal recommendations to menstrual cycle phases
 4. **AI Meal Plan Generator** — A generative AI system that assembles weekly meal plans from local ingredients within user-defined budgets and health constraints
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The Problem

1. The Relevance Gap

Global nutrition apps (MyFitnessPal, Noom, Lose It!) contain fewer than 50 Nigerian foods. A user logging “egusi soup with pounded yam” must approximate it as “vegetable soup with mashed potatoes” — introducing 30-40% calorie estimation error. Nigerian staples like ofada rice, akara, banga soup, and efo riro are entirely absent.

2. The Clinical Blind Spot

Nigeria carries one of the heaviest burdens of diet-related disease in Africa:

- **38% of adults** have hypertension
- **5.77%** have diabetes (projected to double by 2045)
- **10-18% of women** have PCOS, with up to 70% undiagnosed
- **Maternal anemia** affects 67% of pregnant women

Yet no nutrition app offers Nigerian-specific dietary protocols for these conditions. A diabetic Nigerian told to “eat brown rice and quinoa” faces foods that are expensive, unfamiliar, and culturally disconnected.

3. The Gender Health Gap

Menstrual health remains stigmatized, and the connection between cycle phases and nutritional needs is virtually unknown in Nigerian healthcare. Women managing PCOS, preparing for pregnancy, or navigating menopause receive pharmaceutical treatment with minimal dietary support.

4. The Fitness Nutrition Void

Nigeria’s sports nutrition market is valued at \$1.2 billion with 1.5 million gym memberships, yet body-builders import expensive whey protein because no app tells them that moi moi (steamed bean pudding) delivers 15g protein per serving, or that egusi soup with goat meat is a legitimate post-workout meal.

The Solution

NutriAI closes every gap through a single, intelligent platform.

Core Features

Feature	Description	User Benefit
Nigerian Food Database	2,000+ verified local foods with full macro/micro profiles, preparation variants (boiled vs fried plantain), and regional tags	Log meals accurately without approximating
AI Meal Plan Generator	Generates 7-day meal plans from local ingredients matching calorie, macro, and budget targets	Eliminates decision fatigue; ensures nutritional adequacy
Cycle-Synced Nutrition	Adapts recommendations to menstrual phase: iron boost (menstrual), energy optimization (follicular), anti-inflammatory (ovulation), magnesium support (luteal)	First African app connecting hormonal health to culturally appropriate nutrition
Clinical Protocols	Evidence-based dietary programs for PCOS, diabetes, hypertension, pregnancy (trimester-specific), and weight management	Replaces generic advice with condition-specific, locally adapted guidance
Smart Substitute Engine	Suggests culturally acceptable swaps (“Replace eba with amala for lower glycemic load”)	Maintains food enjoyment while meeting health goals
Budget Optimizer	Calculates cost-per-meal and suggests affordable alternatives within a weekly food budget	Makes healthy eating financially sustainable
Snap & Scan	AI-powered food photo recognition trained on Nigerian dishes	Instant calorie and macro logging from a photo
Family Meal Planner	Multi-profile household planning with portion scaling and shared shopping lists	Simplifies nutrition for families of all sizes

Table 1 – continued

Feature	Description	User Benefit
Water & Micronutrient Tracker	Tracks hydration, iron, calcium, folate, and other critical nutrients with Nigerian context	Prevents common deficiencies (anemia, vitamin D)
AI Nutritionist Chat	Conversational AI providing real-time dietary guidance in English, Pidgin, Yoruba, Igbo, and Hausa	Removes language and access barriers to professional nutrition advice

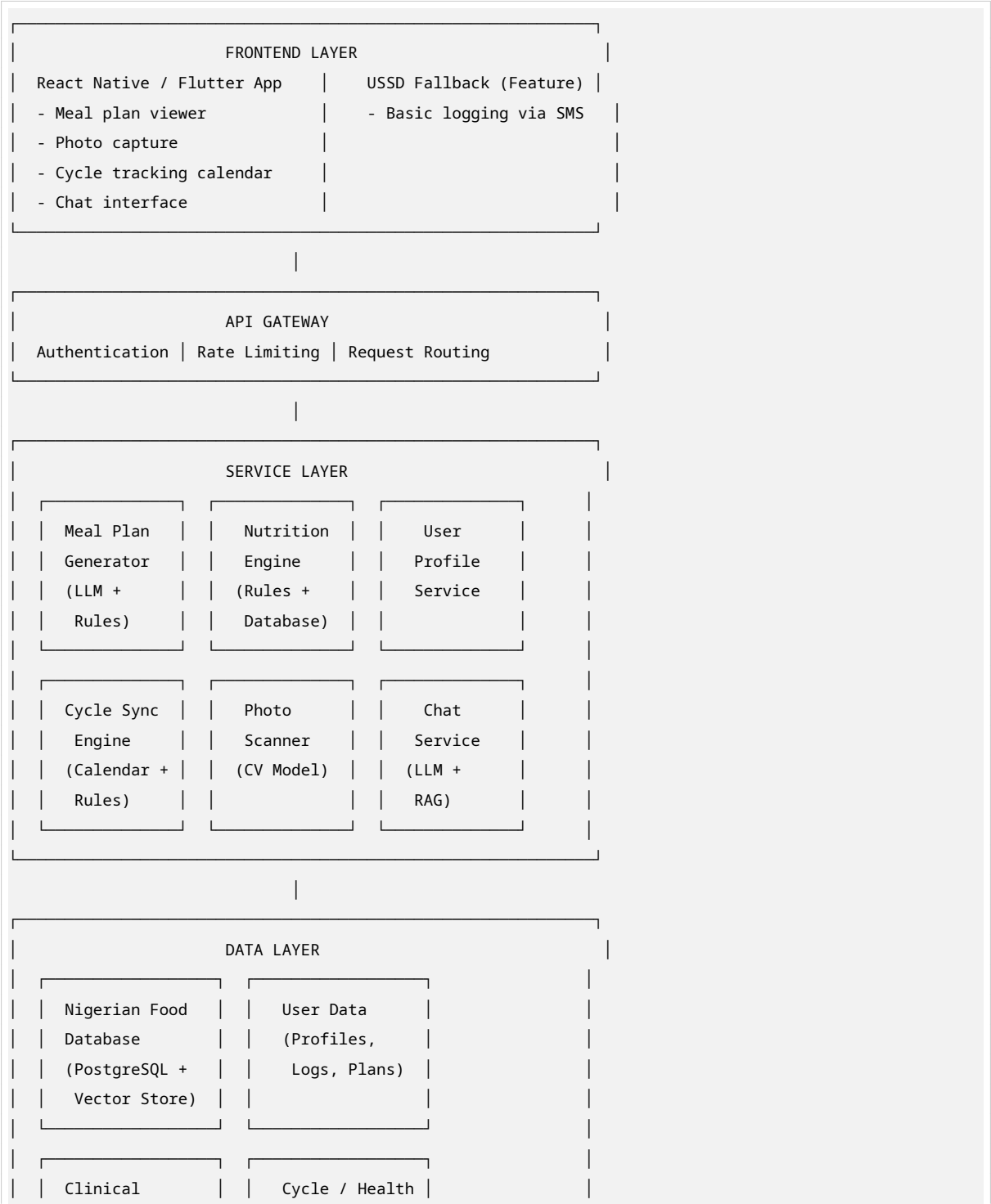
User Segments Served

Segment	Estimated Nigerian Population	Primary Need	NutriAI Adaptation
Menstrual Cycle Sync	25M women	Hormonal nutrition support	4-phase meal recommendations with iron, magnesium, B-vitamin optimization
Students / Budget-Conscious	20M	Affordable healthy eating	Cost-per-calorie optimization; bulk meal prep suggestions
Hypertension Management	12M adults	Sodium control	Low-sodium Nigerian recipes; maggi cube alternatives; DASH-adapted protocols
PCOS Management	8M women	Insulin sensitivity	Low-glycemic Nigerian meal plans; anti-inflammatory protocols
Weight Loss	8M	Caloric deficit	Portion-controlled swallows; vegetable-forward soups; satiety optimization
Pregnancy	4M women annually	Folate, iron, DHA, calcium	Trimester-specific plans; tribe-aware cultural foods; anemia prevention
Postpartum / Lactation	3M women annually	Caloric surplus, hydration	Lactation-supporting foods (pap/ogi, millet, groundnut soup); hydration tracking
Diabetes	5M	Glycemic control	Glycemic load calculator; unripe plantain, beans, portion-controlled swallow protocols
Muscle Gain / Bulking	3M	Protein surplus	Protein timing around Nigerian meals; suya, moi moi, egusi optimization

Table 2 – continued

Segment	Estimated Nigerian Population	Primary Need	NutriAI Adaptation
General Wellness	15M	Balanced nutrition	Standard macro-balanced meal plans with Nigerian variety

Technical Architecture





Key Technical Components

Component	Technology	Purpose
Food Recognition Model	TensorFlow Lite / ONNX	On-device inference for photo scanning; trained on 200 Nigerian dish categories
Meal Plan LLM	GPT-4 / Claude API	Natural language meal plan generation with Nigerian food constraints
Nutrition Rules Engine	Python (Pandas + Custom)	Macro/micro validation against clinical guidelines
Cycle Sync Engine	Calendar API + Static Rules	Phase detection and nutrient recommendation mapping
Chat Service	RAG (Retrieval-Augmented Generation)	Grounded nutrition advice from verified clinical sources
Database	PostgreSQL + Redis	Structured food data with vector search for similarity matching
Mobile App	Flutter	Cross-platform iOS/Android with offline capability

The Unique Angle: Cycle-Synced Nigerian Nutrition

NutriAI is the first nutrition application globally to connect menstrual cycle biology with African dietary patterns. The feature operates across four phases:

Menstrual Phase (Days 1-5):

- Nutritional Focus: Iron replenishment, warmth, comfort
- Nigerian Recommendations: Ugu soup (iron-rich leafy greens), beef suya (heme iron), warm zobo drink, palm oil-free vegetable soups
- Clinical Rationale: Average iron loss of 30-40mg per cycle; Nigerian women have high baseline anemia rates

Follicular Phase (Days 6-14):

- Nutritional Focus: Energy optimization, B-vitamins, lean protein
- Nigerian Recommendations: Moi moi (B-vitamins, protein), grilled catfish, ofada rice, fresh vegetable salads
- Clinical Rationale: Rising estrogen increases insulin sensitivity; optimal window for complex carbohydrates

Ovulation (Day 14):

- Nutritional Focus: Anti-inflammatory support, omega-3s, hydration
- Nigerian Recommendations: Smoked titus (mackerel, omega-3), vegetable salad with sesame seeds, light pepper soups
- Clinical Rationale: Prostaglandin production peaks; anti-inflammatory foods reduce cramping severity

Luteal Phase (Days 15-28):

- Nutritional Focus: Magnesium, complex carbs, mood support
- Nigerian Recommendations: Plantain (magnesium, complex carbs), groundnut soup (B-vitamins), pap with milk (tryptophan for serotonin), dates
- Clinical Rationale: Progesterone rise increases metabolic rate by 5-10%; magnesium deficiency correlates with PMS severity

This feature addresses a genuine clinical gap: PCOS affects 10-18% of Nigerian women, with insulin resistance as a core mechanism. Cycle-synced nutrition that naturally modulates glycemic load through the month is a non-pharmaceutical intervention with strong emerging evidence.

Clinical Validation Strategy

NutriAI's protocols are designed for clinical partnership and validation:

Protocol	Clinical Partner Target	Validation Metric
PCOS Nutrition	Society of Gynaecology and Obstetrics of Nigeria (SOGON)	Fasting insulin reduction at 12 weeks
Diabetes Management	Diabetes Association of Nigeria	HbA1c improvement at 6 months
Hypertension	Nigerian Hypertension Society	24-hour ambulatory BP reduction
Pregnancy	National Primary Health Care Development Agency	Maternal anemia incidence reduction
General Wellness	Registered Dietitians Association of Nigeria	Diet quality score improvement (HEI-Nigeria)

Business Model

Revenue Stream	Description	Timeline
Freemium Subscriptions	Basic meal planning free; premium features (disease protocols, cycle sync, advanced analytics) at ₦1,500/month	Launch

Table 5 – continued

Revenue Stream	Description	Timeline
B2B Clinic Partnerships	White-label deployment in gynecology and endocrinology clinics; revenue share on patient subscriptions	Month 6
Gym & Fitness Partnerships	Branded nutrition layer for fitness centers; bulk subscription pricing	Month 6
Corporate Wellness	Employer-sponsored nutrition programs for staff health benefits	Year 2
Anonymized Nutrition Insights	Aggregated dietary trend data for FMCG companies and public health agencies	Year 2

Competitive Landscape

Dimension	MyFitnessPal	HealthifyMe	Noom	NutriAI
Nigerian Foods	< 50 items	< 20 items	0 items	2,000+ verified
Local Language Support	English only	English, Hindi	English only	English, Pidgin, Yoruba, Igbo, Hausa
Cycle-Synced Nutrition	✗ None	✗ None	✗ None	✓ 4-phase system
PCOS Protocol	✗ None	⚠ Generic	✗ None	✓ Nigerian-specific
Pregnancy (Trimester)	⚠ Generic	⚠ Generic	✗ None	✓ Trimester + tribe-aware
Photo Recognition	✓ Good	✓ Good	⚠ Basic	✓ Trained on Nigerian dishes
Budget Awareness	✗ None	⚠ Basic	✗ None	✓ Cost-per-meal optimizer
Family Meal Planning	⚠ Basic	✗ None	✗ None	✓ Multi-profile household
Gym/Bulking Support	✓ Good	⚠ Basic	✗ None	✓ Nigerian protein sources
Clinical Validation	✗ None	⚠ Some (India)	✗ None	✓ Nigerian medical partnerships

Impact Metrics (12-Month Targets)

Metric	Target	Measurement Method
Active Users	50,000	Monthly active users logging 3 meals/week
Meal Plans Generated	500,000	Cumulative AI-generated weekly plans
Food Log Accuracy	85%	User-reported portion accuracy vs. photo scan
Clinical Protocol Adherence	70%	Users completing 4+ weeks of disease-specific plans
Diet Quality Improvement	+15% HEI-Nigeria score	Pre/post survey of active users
Anemia Awareness	10,000 women screened	Self-reported iron intake vs. RDA
Partner Clinics	25	Active B2B deployments

Why This Wins

- Cultural Authenticity:** Built for egusi, not quinoa. Nigerian food is the default, not an afterthought.
- Clinical Credibility:** Evidence-based protocols adapted for Nigerian disease epidemiology, not Western templates.
- Gender Intelligence:** The first African nutrition app to treat menstrual health as a nutrition science, not a taboo.
- Economic Accessibility:** Budget-aware meal planning makes healthy eating viable for students and low-income households.
- Technical Feasibility:** Core features (meal generator, tracker, cycle sync) are buildable with existing AI/ML tools; photo scanner is a scalable enhancement.
- Defensible Data Asset:** The Nigerian food composition database grows more valuable with every user interaction, creating a compounding competitive moat.

Team & Execution

Hackathon MVP Scope (48 Hours):

- Nigerian food database: 50 core staples with macros
- User onboarding: Profile creation with health conditions and goals
- AI meal plan generator: 7-day plans for 3 user segments (PCOS, pregnancy, muscle gain)
- Cycle sync demo: Static phase-based recommendations
- Basic calorie tracker: Manual entry with food search

Post-Hackathon Roadmap:

- Month 1-2: Expand database to 500 foods; beta test with 100 users
- Month 3-4: Launch photo scanner (v1); integrate cycle tracking calendar
- Month 5-6: Clinical pilot with 3 partner clinics; publish validation protocol

- Month 7-12: Scale to 50,000 users; launch B2B partnerships

NutriAI — Because eating well shouldn't require importing quinoa.